

2025 Lab report 1: Due Friday March 7th at 10 pm Orlando time!

We are using online submission of your lab reports. Please submit each report **as a pdf file**. Also check the file size and try to reduce the size using “reduce size” or “optimize” options as you save the file so that its <10 MB.

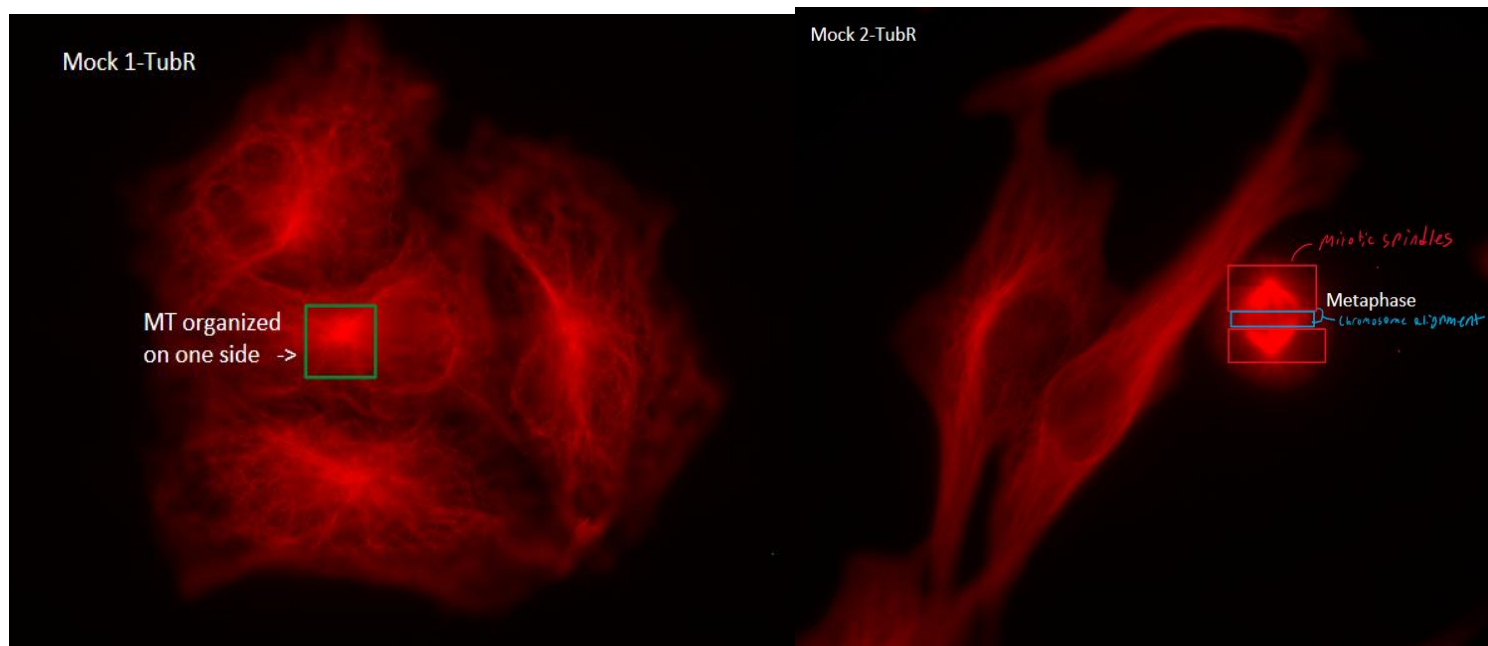
Lab Report 01: Cells on Drugs: Taxol

In lab in on February 4th, we used Taxol to examine what happens to the cytoskeleton after treatment. The images you need to use to answer the lab report questions have been copied into the **Lab Report** folder. **NOTE:** Please use the posted pictures that correspond to files identified in the Lab Report, even if you find other groovy cool pictures.

1) Make sure you save these files for this lab report with a file name that starts with your first name. Also, put your name either as a header on the report or on the first page of the report (2 points)

Taxol questions:

2) Microtubules in control cells: look at the Mock1-TubR.jpg and Mock2-TubR.jpg files. **Post images of the microtubule network from mock interphase cells and mock mitotic cells. (10 points)**



Explain what differences (if any) you see in the microtubule cytoskeleton between the interphase from the Mock1 file and mitotic cells from the mock2 file. **(10 points)**

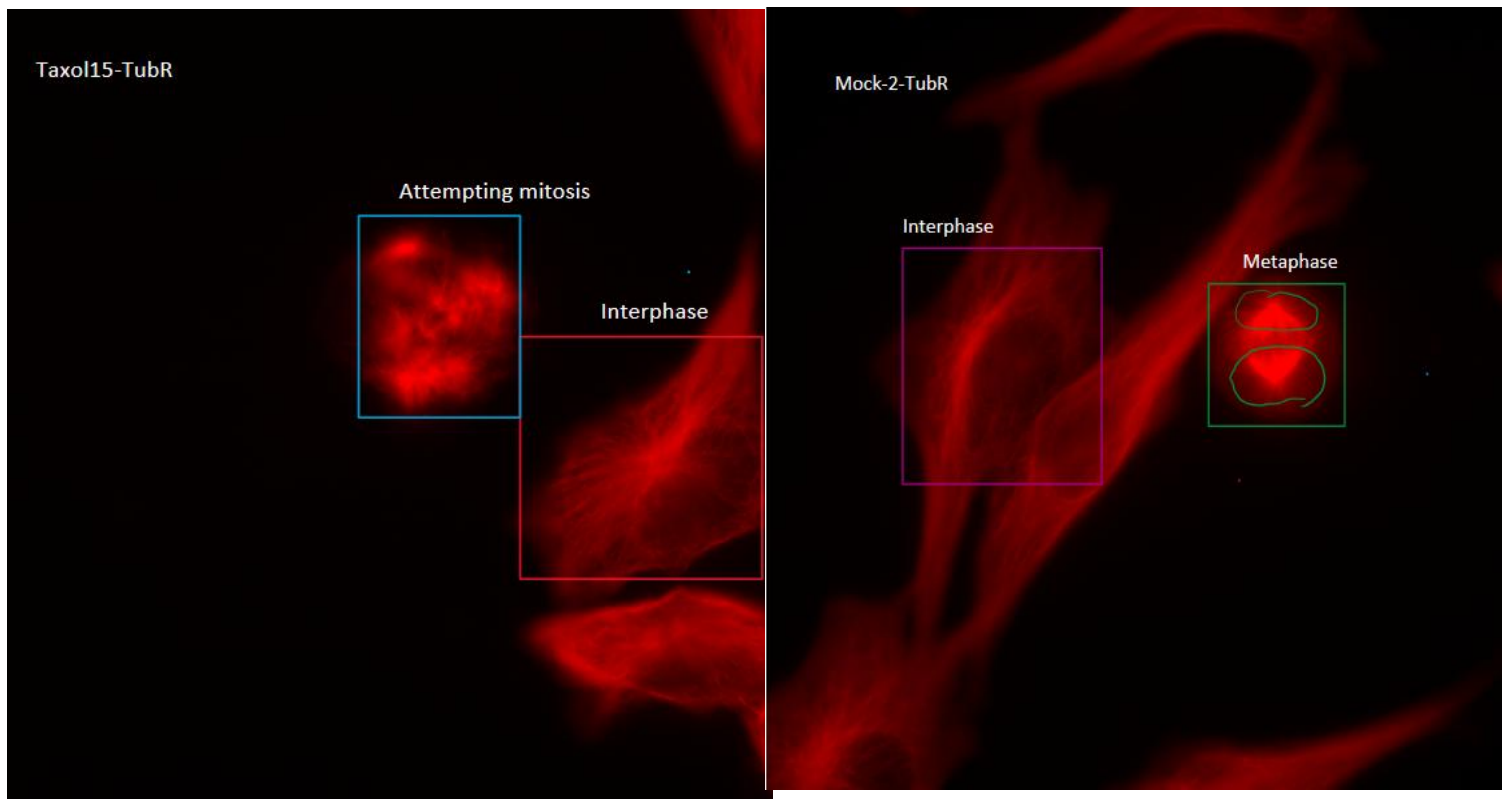
In the interphase picture we can see tubulin arranging itself on one side of the cell (shown by the green box). It creates a large and fairly unorganized plate of spaghetti tubulin surrounding the cell for structural support and creating tracks for motor proteins to transport various materials across the cell. The tubulin is bound to a microtubule organizing center which is why there is a high concentration of tubulin originating from one spot.

Claire Hiscock

The mitotic cell is in metaphase. As outlined in the red box there is a high concentration of microtubules which are forming mitotic spindles. Cross-referencing this photo with the DAPI stain shows a high concentration of blue in the center of the cell which is dark in these photos (blue box), the mitotic spindles are attaching to the chromosomes in the center of the cell. I originally thought that the cell immediately left of the metaphase cell was starting telophase because its tubulin is so stretched out however, it appears that RD cells have a more stretched structure than other cell types.

On the images, label specific regions or areas that best show any differences. **(5 points)**
Green, red, and blue boxes referenced in the description above

3) Microtubules in Taxol treated mitotic cells: look at the Taxol15-TubR.jpg file. Post images of the microtubule network from taxol-treated and mock2 mitotic cells. **(10 points)**



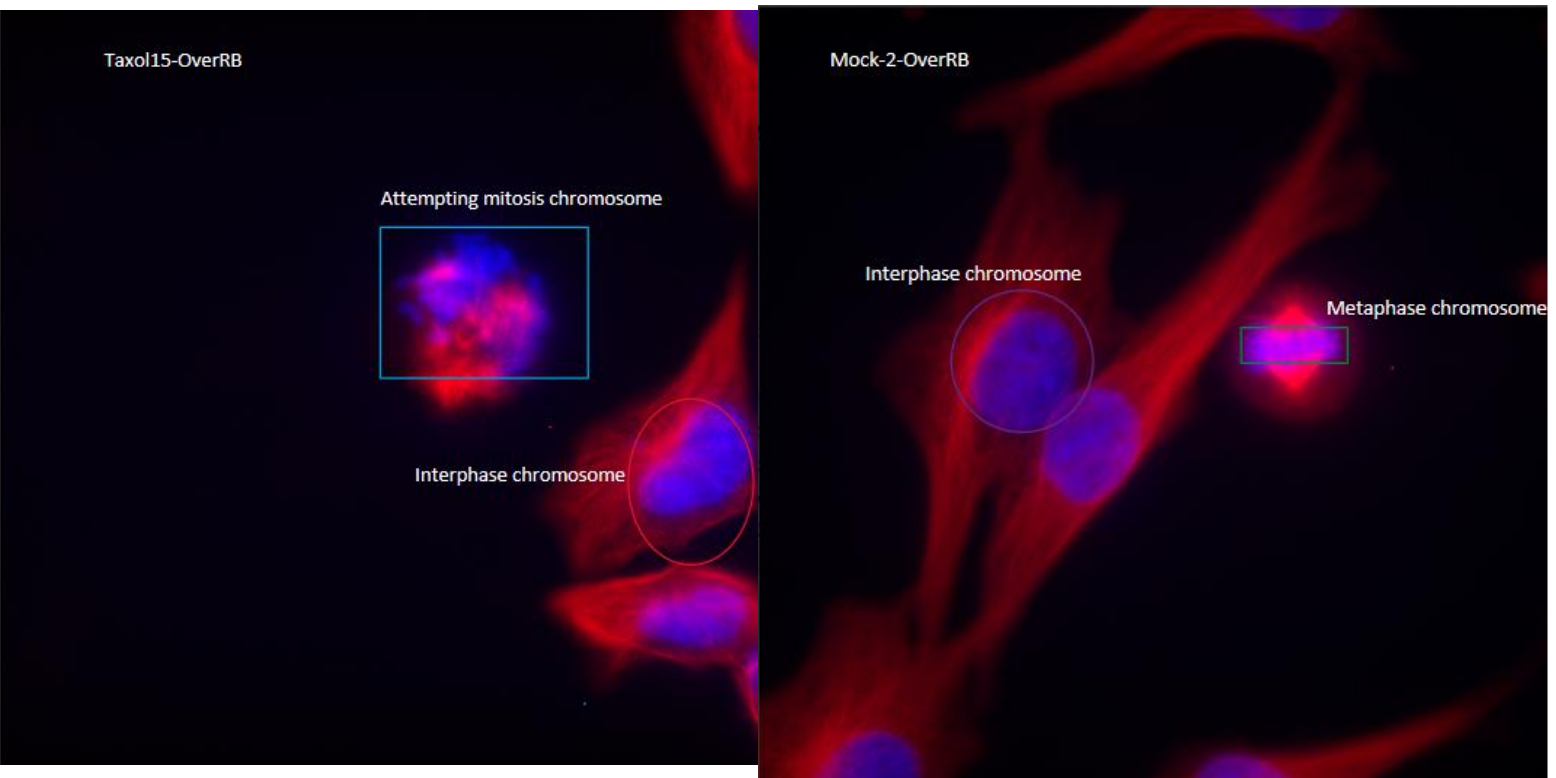
Explain what differences (if any) you see in the microtubule cytoskeleton between the Taxol-treated and mock **mitotic cells**. **(10 points)**

For the cells in interphase (outlined in red[Taxol] and purple[mock]), there is not a very dramatic difference between the two's microtubules. Both show a high microtubule concentration on one side of the cell and creating a network which outlines the cell. However, the Taxol condition appears to be more diffused than the mock cells.

Conversely, the control cell in metaphase (outlined in green) and the Taxol-treated cell attempting mitosis (outlined in blue) have stark contrasts. The Taxol-treated cell's microtubules are not organized at 2 opposite poles in the same way as the control cell's (circled in green). Instead, the microtubules appear scattered around randomly. The cell treated with Taxol cannot catastrophize its microtubules as normal. This means that the mitotic spindles cannot form properly therefore, chromosomes cannot be aligned, and the cell cannot successfully complete mitosis.

On the images, label specific regions or areas that best show any differences. **(5 points)**
Red, purple, blue, green boxes and circles as referenced in the above question.

4) Now let's look at **spindle organization**. Post pictures from Taxol15-OverRB.jpg and Mock2-OverRB.jpg. **(5 points)**



Look at the way the chromosomes interact with the microtubule spindles. What difference do you see in how mitotic spindles are organized in mock versus taxol-treated cells? **(10 points)**

The chromosomes of the metaphase control cell and the cell attempting mitosis in the Taxol-treated condition are drastically different. In the metaphase mock cell the chromosomes are neatly lined up in the center of the cell (outlined in green) and the mitotic spindles appear to overlap with the chromosome indicating that they are bound. However, in the Taxol treated cell attempting mitosis the chromosomes (outlined in blue) are not neatly connected on both sides by the chromosomes. Instead, the chromosome is unequally dispersed towards one side of the cell and some parts of it do not appear to be connected to microtubules.

Claire Hiscock

The Taxol-treated cell's chromosomes (outlined in red) in interphase create more of a bean shape rather than an oval in the control cells (outlined in purple). Microtubules are necessary for establishing polarity of the cell, the Taxol-treated cell can no longer catastrophize it's microtubules leading to an unwanted polarity.

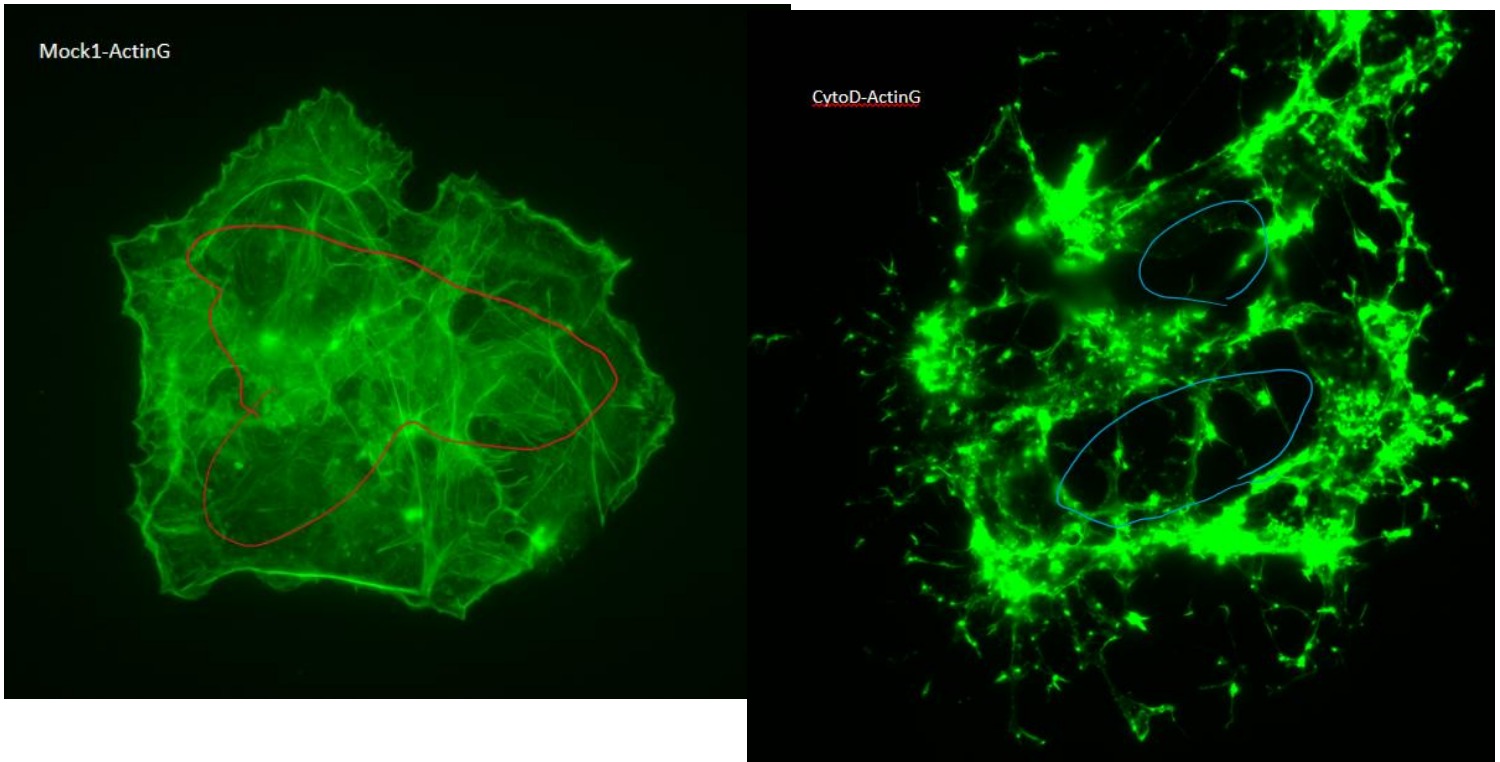
On the images, label specific regions or areas that best show any differences. **(5 points)**
Red, blue, green, and purple outlines referenced above.

5) Before we did the actual experiment, what expectation did you have for what the drug taxol would do to the cell cytoskeleton? How close was your expectation to what we saw, especially in mitotic taxol-treated cells? **(8 points)**

My expectation of Taxol was that it would bind to and stabilize the microtubules of the cell leading to never-ending growth and eventual apoptosis of the cell. Taxol does lead to cell death but there was no never-ending growth of the microtubule. I also thought that the cell would not try to undergo mitosis and simply die but in the real experiment the cells still attempt to divide. My expectation was fairly far-off of what really happened, but I now know a lot more about microtubules!

6) **Cytochalasin D questions:**

2) **Actin in CytoD treated cells:** look at the Mock1-ActinG.jpg and CytoD-ActinG.jpg files. **Post images of the action network from CytoD treated and mock cells.** (10 points)



Explain what differences (if any) you see in the actin cytoskeleton between the CytoD treated and mock cells. (10 points)

In the control mock cells, the actin cytoskeleton appears almost continuous across 4 different cells (cross referenced with DAPI stain, red outline refers to the actin skeleton around the rough placement of the chromosomes). However, the CytoD condition cells have large gaps across the actin cytoskeleton (outlined in blue). Cytochalasin D binds to the plus end of an actin filament and prevents it from adding more actin. In the CytoD treated condition, the actin network disassembled leading to large gaps around the cell and destroying the interconnected network of actin.

On the images, label specific regions or areas that best show any differences. (5 points)
Red and blue outlines referenced above.

5) **Before we did the actual experiment**, what expectation did you have for what the drug cytochalasin D would do to the cell cytoskeleton? How close was your expectation to what we saw? (5 points)

I expected that cytochalasin D would lead to big gaps in the actin cytoskeleton which was exactly what happened. However, I did not realize how interconnected the actin network is in normal cells and that treatment of CytoD would dismantle a network across multiple cells.

